

Water quality analysis

1140 Plett Rd, Cadillac MI

2025 vs 2026 PFAS comparison

Year-over-year findings, EPA Hazard Index calculation, and prioritized risk assessment

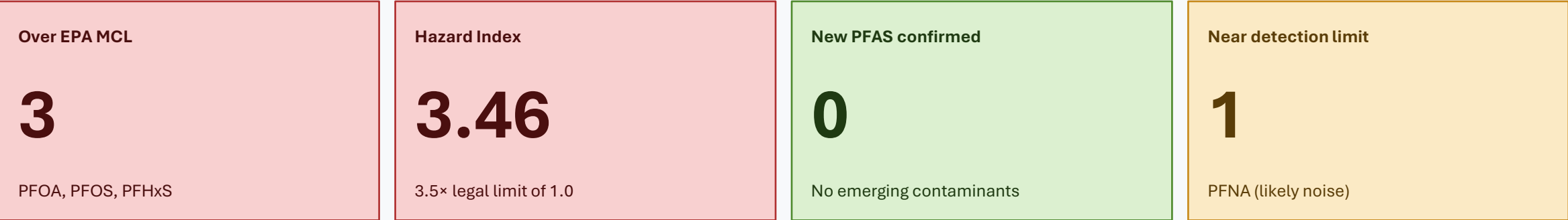
Prepared April 2026

Status key

Good
Caution
High concern
Over legal limit

Executive summary

Three independent EPA failures. No emerging contaminants. Persistence is the story.



Key takeaways

- **Persistent contamination.** PFOA (34 → 34.6 ppt), PFHxS (36 → 33.4 ppt), and PFOS (19 → 10.6 ppt) all remained above MCL across seven months — consistent with a stable groundwater source.
- **Triple-failure profile.** Water fails EPA standards on three independent grounds: PFOA individual MCL, PFOS individual MCL, and PFAS mixture Hazard Index.
- **No new PFAS detected.** PFNA near-RL estimate (1.1 ppt) is most likely detection-limit noise, not emerging contamination.
- **Treatment priority: PFHxS.** PFHxS alone drives 96% of the Hazard Index. Certified activated carbon or RO systems address all three violating compounds.

Data scope and standardization

Why only PFAS can be fairly compared year-over-year.

2025 dataset	2026 dataset
Source: EGLE / MPART PFAS Comparability Study Dates: Sep 10, 2025 (plus Cyclopure Mar/Jun 2025) Labs: EGLE, Merit Labs, Cyclopure (3 parallel) Method: EPA 533 + Cyclopure proprietary Panel: 25 to 55 PFAS compounds Units: ng/L (= ppt)	Source: Tap Score report QWE6BQ (SimpleLab) Date: Apr 9, 2026 — Break Room Sink Labs: Microbac, Alliance Technical, EMSL, TG Analytical Method: EPA 533, 524.2, 200.7/.8, 300.0, 900.0 Panel: PFAS + metals + VOCs + anions + radiation + PPCPs + SVOCs + aldehydes Units: ppb and ppm (converted to ppt for PFAS)

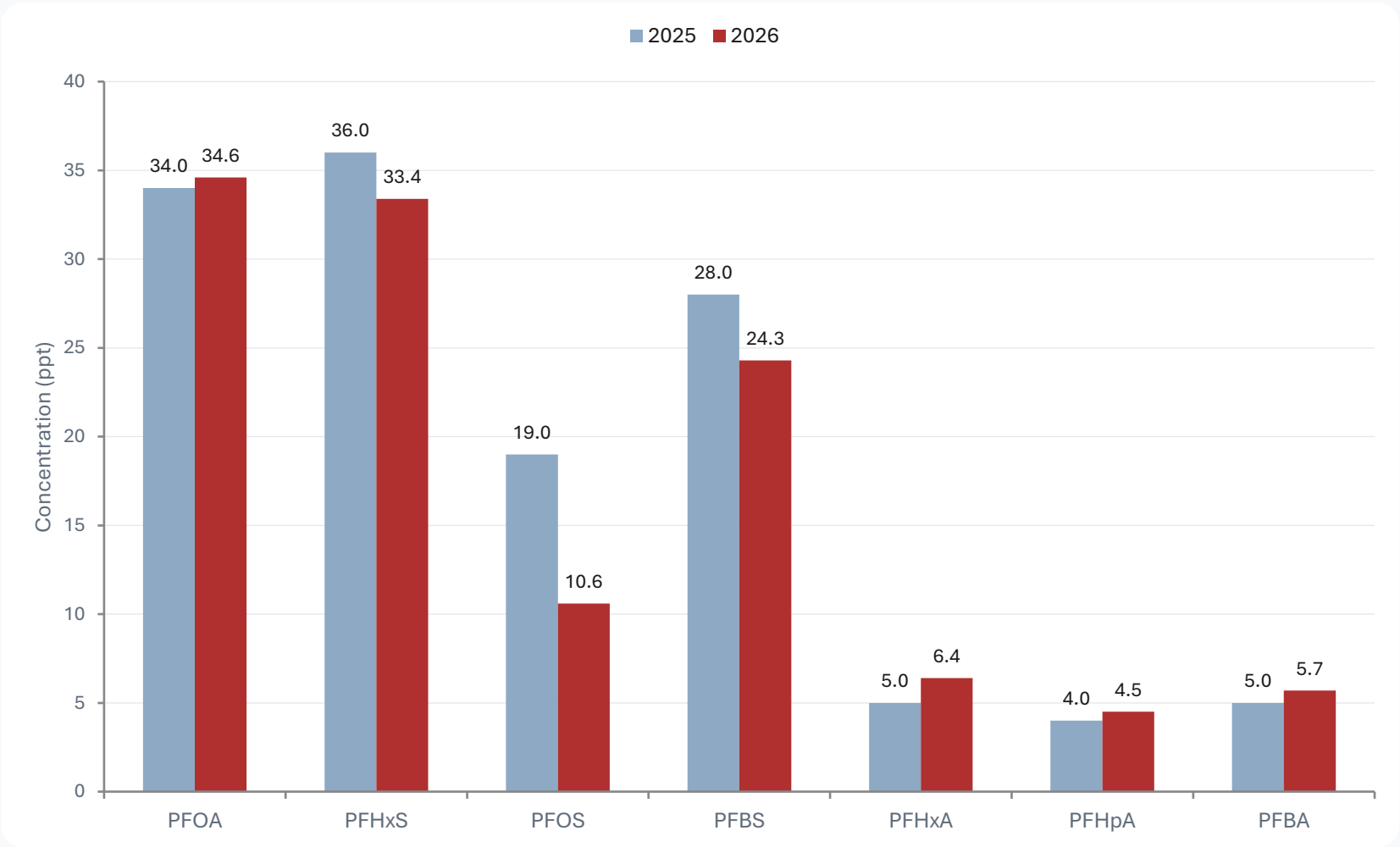
Data-scope limitation

The 2025 dataset tested only PFAS. The 2026 dataset tested PFAS plus metals, VOCs, anions, and radiation. Any non-PFAS analyte detected in 2026 has no 2025 baseline — it cannot be called "emerging" because it was never measured the year before. Only PFAS supports a valid year-over-year comparison.

Standardization applied: name alignment by CAS number, unit conversion (1 ppb = 1,000 ppt), primary comparator is EGLE indoor unflushed (same EPA 533 method as 2026).

PFAS concentrations: 2025 vs 2026

Compounds ranked by MCL exceedance. Red panels mark levels over the EPA enforceable limit.



EPA enforceable limits

PFOA Result: 34.6 ppt MCL: 4 ppt 8.6× over
PFHxS Result: 33.4 ppt MCL: 10 ppt 3.3× over
PFOS Result: 10.6 ppt MCL: 4 ppt 2.6× over
PFBS Result: 24.3 ppt MCL: 2,000 ppt well below

PFBS uses a health-based value rather than an individual MCL.

Full PFAS comparison table

All values in ppt (ng/L). The 2025 column is the EGLE lab indoor unflushed sample.

Analyte	2025 (ppt)	2026 (ppt)	EPA MCL (ppt)	Status
PFOA	34	34.6	4	Over legal limit
PFHxS	36	33.4	10	Over legal limit
PFOS	19	10.6	4	Over legal limit
PFBS	28	24.3	2,000 HBV	Below HBV
PFHxA	5	6.4	—	High concern
PFHpA	4	4.5	—	Good
PFBA	5	5.7	—	Good
PFPeA	4	3.5	—	Good
PFNA	ND	1.1*	10	Caution (near RL)
PFHpS	ND	1.1*	—	Caution (near RL)
PFPeS	2	1.5*	—	Caution (near RL)
GenX (HFPO-DA)	ND	ND	10	Good
13 other PFAS	ND	ND	varies	Good

* value estimated below the reporting limit (RL = 1.9 ppt for Tap Score, 2 ppt for EGLE). Interpret as qualitative detection only.

EPA PFAS Hazard Index

The mixture rule: when multiple PFAS co-occur, the sum of health ratios must stay below 1.0.

How it works

$$HI = [PFHxS]/10 + [PFNA]/10 + [GenX]/10 + [PFBS]/2,000$$

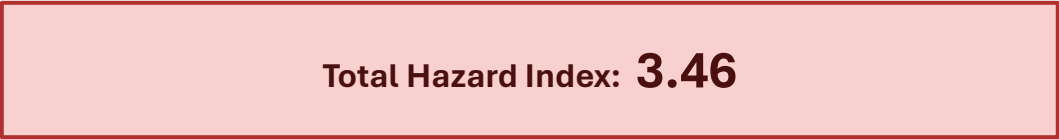
For each of four co-occurring PFAS, divide the measured concentration by its Health-Based Water Concentration (HBWC), then sum the ratios.

HI > 1.0 = MCL violation

PFOA and PFOS are not in the HI — each has its own 4 ppt individual MCL and is evaluated separately.

Applied to 2026 values

Compound	Value (ppt)	HBWC	Ratio
PFHxS	33.4	10	3.34
PFNA (est.)	1.11	10	0.11
GenX (HFPO-DA)	ND	10	0.00
PFBS	24.3	2,000	0.012



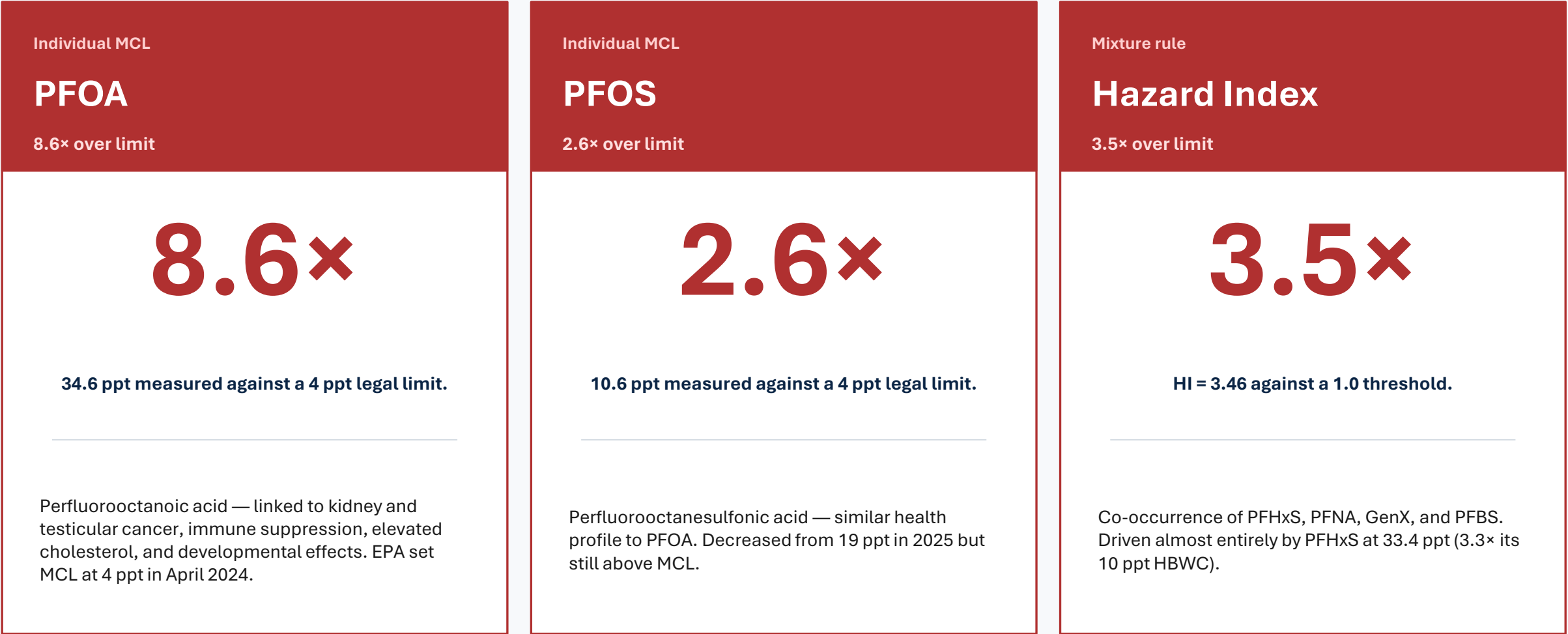
How far over the 1.0 legal limit



PFHxS alone contributes 3.34 of the 3.46 — roughly 96% of the total risk.

Three independent EPA failures

The water fails federal drinking water standards on three separate grounds.



Health significance and priorities

Ranked by severity and certainty of the evidence.

HIGH	PFOA, PFOS, PFHxS continue to exceed EPA MCLs Linked to kidney and testicular cancer, immune suppression, developmental effects, and elevated cholesterol. MCLs are set near the limit of detection because health effects occur at very low doses. Persistence across seven months indicates an ongoing source, not a transient event.
HIGH	No evidence of PFAS source remediation PFOA held at 34 → 34.6 ppt; PFHxS at 36 → 33.4 ppt. Consistency across sampling dates points to a stable, continuous groundwater plume. Waiting for natural attenuation is not a viable strategy.
MEDIUM	PFOS apparent decrease (19 → 10.6 ppt) Could reflect real variability, different sampling conditions (unflushed vs. flushed, different fixture), or measurement uncertainty. Still above MCL — not actionable as improvement.
LOW	PFNA estimated at 1.1 ppt below reporting limit Below reporting limits in both years; likely within detection noise. Worth retesting but not currently a confirmed change.
N/A	All non-PFAS analytes in 2026 No 2025 baseline exists — metals, VOCs, anions, and radiation were not measured in 2025. Cannot be characterized as emerging until a baseline panel is established.

2026 broader panel — for reference only

No 2025 baseline exists for these analytes. Values shown for situational awareness, not change detection.

Manganese 0.132 ppm Above 0.05 ppm aesthetic secondary limit	Nitrate (as N) 3.9 ppm Elevated for a well; below 10 ppm MCL	Lead 1.66 ppb Below 15 ppb action level
Arsenic 0.64 ppb Well below 10 ppb MCL	Copper 0.087 ppm Well below 1.3 ppm action level	Chromium (total) 1.1 ppb Well below 100 ppb MCL
Gross Beta radiation 3.91 pCi/L Below 50 pCi/L screening	Gross Alpha radiation Below MDL Non-detect	All VOCs (EPA 524.2) Not detected About 60 volatile organics tested
Total THMs Not detected Trihalomethanes below RL	Tannins Not detected Below 0.3 ppm RL	pH 7.7 Within 6.5 to 8.5 secondary range

Pharmaceuticals and personal care products

Across two panels covering 83 compounds, 8 were detected. Individual concentrations are low, but the pattern is a fingerprint of wastewater influence.

Compounds tested 83 Across two PPCP panels	Detected 8 Six quantified, two estimated	With EPA MCL 0 No enforceable limits exist	Signature Wastewater Sweeteners + pharma + antibiotics
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Detected compounds by class

Artificial sweeteners Wastewater tracers Sucralose 1,100 ppt Acesulfame-K 54 ppt	Pharmaceuticals Anticonvulsants and anxiolytic Carbamazepine 16 ppt anticonvulsant Primidone 16 ppt anticonvulsant Meprobamate 6.1 ppt anxiolytic	Antibiotics Includes veterinary drugs Sulphadiazine 8.1 ppt Sulfadimethoxine 4.7 ppt* veterinary Sulfamethazine 3 ppt* veterinary
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Interpretation Sucralose and acesulfame-K pass through humans and conventional wastewater treatment largely unchanged — their presence is a strong indicator of septic or sewage influence on the groundwater source. Veterinary antibiotics additionally suggest agricultural runoff. Individual concentrations are low, but the combined signature warrants source investigation.

SVOCs and aldehydes

Semivolatile organics (62 compounds) and aldehydes (10 compounds) — one trace detection between them.

Semivolatile organics (SVOCs)

62 compounds | 1 trace detection | 61 not detected

DEHP — Bis(2-ethylhexyl) phthalate

1.36 ppb*

EPA MCL 6 ppb · your value ~23% of MCL, estimated below reporting limit (qualitative only).
Phthalate plasticizer, commonly leaches from PVC plumbing and plastic containers.

All other SVOC classes: not detected

- PAHs (16 compounds)
- Chlorinated benzenes (4)
- Phthalates (5 others)
- Phenols, chlorophenols (10)
- Nitroaromatics (5)
- Other SVOCs (21)

Aldehydes

10 compounds | all not detected

All 10 aldehydes not detected

Aldehydes (including formaldehyde) typically form as disinfection byproducts. Their absence aligns with untreated well water that has not been chlorinated.

Compounds tested

- Acetaldehyde
- Crotonaldehyde
- Formaldehyde
- Hexanal
- Octanal
- Butanal
- Decanal
- Heptanal
- Nonanal
- Propanal

Health impact of detected chemicals

Documented cause-and-effect per compound. Sourced from EPA IRIS, ATSDR toxicological profiles, and peer-reviewed epidemiology.

PFOA 34.6 ppt Perfluorooctanoic acid 8.6× MCL - Kidney cancer - Testicular cancer - Reduced vaccine response in children - Lower infant birth weight	PFOS 10.6 ppt Perfluorooctanesulfonic acid 2.6× MCL - Liver damage - Thyroid dysfunction - Pregnancy-induced hypertension - Neurodevelopmental effects in children	PFHxS 33.4 ppt Perfluorohexanesulfonic acid 3.3× HBWC - Thyroid disease - Liver toxicity - Immune suppression - Developmental delays in children
Manganese 0.132 ppm Above 0.05 ppm secondary limit 2.6× aesthetic - Parkinsonism-like neurotoxicity at chronic exposure - Cognitive and behavioral changes in adults - Infant brain development harm - Bottle-fed babies at highest risk	DEHP phthalate 1.36 ppb* Bis(2-ethylhexyl) phthalate Endocrine disruptor - Hormone-signaling disruption - Reduced male fertility and sperm count - Developmental effects in boys in utero - Probable human carcinogen (IARC 2B)	Additional PFAS PFBS, PFHxA, PFBA, PFHpA, +4 more Class effect - Persistent — accumulate in blood indefinitely - Endocrine disruption across the class - Add to Hazard Index burden - Liver and kidney stress
Lead 1.66 ppb Below 15 ppb action No safe level - Permanent IQ loss in children - Learning and behavioral disorders - Cardiovascular disease in adults - Kidney damage, reduced fertility	Nitrate 3.9 ppm Below 10 ppm MCL Elevated - Blue baby syndrome in infants under 6 months - Thyroid interference - Colorectal and gastric cancer at chronic exposure - Pregnancy complications	PPCPs & antibiotics 8 detections Trace pharma + sulfonamides Indicator + AMR - Trace pharma: long-term effects poorly studied - Antibiotics drive environmental AMR - Sweeteners confirm wastewater pathway - Continuous low-dose mixture exposure

Likely contamination sources

How each detected compound probably entered this water supply — with the chemicals each pathway is known to carry.

Firefighting foam (AFFF)

Aqueous film-forming foam used at airports, military bases, refineries, and fire-training sites. Historically the largest single source of PFAS groundwater contamination in the U.S.

Detected in this sample
PFOA, PFOS, PFHxS, other PFAS

Industrial manufacturing

Chemical plants, metal-plating, paper mills, and tanneries that produced or applied PFAS. Plastic and PVC manufacturing releases DEHP phthalate. Michigan has known legacy PFAS industrial sites.

Detected in this sample
All PFAS, DEHP phthalate

Wastewater & septic systems

Conventional treatment plants cannot remove pharmaceuticals, artificial sweeteners, or PFAS. Septic systems provide no treatment at all. Effluent seeps into groundwater.

Detected in this sample
Sucralose, Acesulfame-K, Carbamazepine, Primidone, Meprobamate, Sulphadiazine

Landfill leachate

PFAS-coated consumer goods (cookware, food packaging, textiles, cosmetics) break down in landfills. Plastic waste releases phthalates. Leachate migrates into groundwater.

Detected in this sample
PFAS, DEHP, other phthalates

Agricultural runoff

Fertilizer runoff, animal feedlots, biosolids applied to cropland as soil amendment, and veterinary drug use in livestock operations. Biosolids are a documented PFAS pathway.

Detected in this sample
Nitrate, Sulfadimethoxine, Sulfamethazine

Natural geology & plumbing

Michigan's glacial aquifers naturally contain elevated manganese. Older plumbing corrodes and adds lead; PVC pipes leach trace DEHP over time.

Detected in this sample
Manganese, Lead, trace DEHP